

Parent Functions III: $1/x$ and Asymptotes**TEACHER KEY** | Algebra II Honors | Unit 1 | Day 6 | TEK A2.2.A**Objective:** I can graph the reciprocal parent function, describe its end behavior precisely, prove it is odd, and locate vertical asymptotes of related rules from the denominator.**Vocabulary****Vertical asymptote** — a vertical line the graph **never crosses**, located where the function is **undefined**.**Horizontal asymptote** — describes **end behavior**. Unlike a vertical one, it **may** be crossed by other functions.**End behavior notation** — "as $x \rightarrow \infty$, $f(x) \rightarrow 0$ " is read: as x grows **without bound**, the output approaches 0.**Part 1 — Build the Table**

x	-4	-2	-1	-1/2	0	1/2	1	2	4
f(x) = 1/x	-1/4	-1/2	-1	-2	undef.	2	1	1/2	1/4

Why is $x = 0$ excluded, and what is the domain in set notation?Division by zero is undefined, so **$\{x \mid x \in \mathbb{R}, x \neq 0\}$** **Part 2 — End Behavior, Precisely**

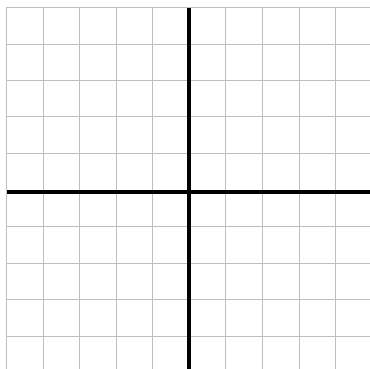
Complete both patterns, then write the behavior in words.

x	1	0.1	0.01	0.001
1/x	1	10	100	1000

x	1	10	100	1000
1/x	1	0.1	0.01	0.001

As $x \rightarrow 0$ from the right, $f(x) \rightarrow +\infty$ As $x \rightarrow 0$ from the left, $f(x) \rightarrow -\infty$ As $x \rightarrow \infty$, $f(x) \rightarrow 0$ As $x \rightarrow -\infty$, $f(x) \rightarrow 0$ So the asymptotes are $x = 0$ and $y = 0$. Does the graph ever reach $y = 0$? **no****Part 3 — Graph and Prove**

Each square is 1 unit. Two branches - do not connect them across the y-axis.

f(x) = 1/x, with both asymptotes dashed.**Prove f is odd.**

$$f(-x) = 1/(-x) = -(1/x)$$

$$\text{so } f(-x) = -f(x)$$

Quadrants used: **I and III**Intercepts: **none**Max or min: **neither**

Why does having no intercepts follow from the asymptotes?

The axes **are** the asymptotes, and a graph never reaches its asymptotes.

Part 4 — Asymptotes of Related Rules

Reason from the denominator. Where it is zero, there is a vertical asymptote.

Function	Denominator = 0 when	Vertical asymptote(s)	Domain
$1/(x - 3)$	$x = 3$	$x = 3$	$(-\infty, 3) \cup (3, \infty)$
$1/(x^2 - 4)$	$x = \pm 2$	$x = 2$ and $x = -2$	$x \neq \pm 2$
$1/(x^2 + 1)$	never	none	all reals

The last row has no vertical asymptote at all. Explain why:

$x^2 + 1$ is **always at least 1**, so it is never zero for a real x .

Part 5 — Going Further

1. True or false, with a reason: a graph may cross a horizontal asymptote. **true**
2. Does $1/x$ itself ever cross $y = 0$? **no** Is that a contradiction with item 1? **no - "may" is not "must"**
3. Solve $1/x = x$. Then say what the solutions mean on the graph.

$x = \mathbf{1 \text{ and } -1}$ — **where $1/x$ meets the line $y = x$**

Exit Ticket

Give the domain and both asymptotes of $f(x) = 1/(x + 5)$, and state its end behavior as $x \rightarrow \infty$.

Domain **$(-\infty, -5) \cup (-5, \infty)$** VA **$x = -5$** HA **$y = 0$**

End behavior: as $x \rightarrow \infty$, $f(x) \rightarrow \mathbf{0}$